

Q.24 Write four methods of making CNT. Write one method in detail.

Q.25 With a neat sketch, write the process of filament winding, using carbon fiber & epoxy resin.

No. of Printed Pages : 4
Roll No.

222263B

6th Sem / Plastic Technology
Subject : Polymer Composites

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 In filament winding process, the fiber is used in the form of

- a) Powder b) Very long fiber
- c) Short fibers d) Very short fibers

Q.2 Which of the following plastics can be reinforced?

- a) PP b) PVC
- c) PC d) All of above

Q.3 Pitch of coal tar Used to make

- a) Glass fiber b) Carbon fiber
- c) Both a & b d) Neither a nor b

Q.4 For aerospace applications, which resin is preferred

- a) Polyester b) Epoxy
- c) PP d) PVC

Q.5 Nano fiber has one dimension in the range of

- a) 10^{-2}m b) 10^{-4}m
- c) 10^{-6}m d) 10^{-9}m

Q.6 Which of the following is laminate?

- a) Table top made by PP
- b) Table top made by PVC
- c) Table top made by Plywood
- d) Table top made by Plane wood

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 FRP is also plastics composite.(T/F)

Q.8 Expand the term PAN.

Q.9 Expand the term PRP.

Q.10 Define E-glass fiber.

Q.11 Which laminate is used to pack potato chips?

Q.12 Define Nano particle.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What is fly ash? How is it produces?

Q.14 What is CNT? Give its two applications.

Q.15 Write four properties of carbon fibers.

Q.16 Write four applications of plastics laminates.

Q.17 What is difference between inert filler & reactive filler?

Q.18 How glass fiber is prepared.

Q.19 Write four properties of epoxy resins.

Q.20 What is difference between S-glass fiber, C-glass fiber & E-Glass fiber?

Q.21 Write the name of four metal particles used in plastics composites.

Q.22 Write a typical formulation for making glass fiber reinforced polymer.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Write the various methods of making laminates. With a neat sketch, explain any one method in detail.